

Problem 7

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Problem 1 The function f is defined by $f(x) = \frac{6e^x + 1}{3e^x + 5}$.

- (a) Find all vertical asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Since the exponential function, e^x , is always positive, it follows that the denominator of the function f is also always positive. Since the exponential function, e^x , is continuous on $(-\infty, \infty)$, it follows that the functions $6e^x + 1$ and $3e^x + 5$ are also continuous on $(-\infty, \infty)$. Therefore, the function f is also continuous there, since it is a quotient of two continuous functions, and the function in the denominator is never 0.

Therefore, f has no vertical asymptotes, since it is continuous on $(-\infty, \infty)$.

- (b) Find all horizontal asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

$\lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5} &= \lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5} \cdot \frac{\frac{1}{e^x}}{\frac{1}{e^x}} = \\ &= \lim_{x \rightarrow \infty} \frac{6 + \frac{1}{e^x}}{3 + \frac{5}{e^x}} = \frac{6 + 0}{3 + 0} = 2 \end{aligned}$$

The line $y = 2$ is a horizontal asymptote of f because $\lim_{x \rightarrow \infty} f(x) = 2$.

Check End Behavior as $x \rightarrow -\infty$:

$$\lim_{x \rightarrow -\infty} \frac{6e^x + 1}{3e^x + 5} = \frac{0 + 1}{0 + 5} = \frac{1}{5}$$

The line $y = \frac{1}{5}$ is a horizontal asymptote of f because $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{5}$.